

Cartwheel Homework 2

Authenticated endpoint and tracing report

6 live traced requests	2 selected trace records	2 prompt versions compared	136 offline tests passed
---------------------------	-----------------------------	-------------------------------	-----------------------------

Outcome

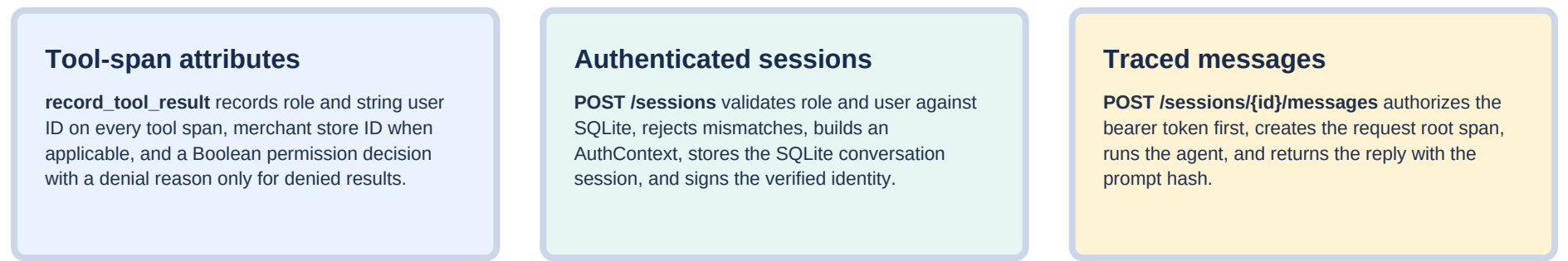
The Cartwheel agent now exposes authenticated HTTP sessions and emits a complete request trace containing root, agent, model, and tool spans. Live evidence confirms identity propagation, permission decisions, message capture, model and token metadata, and prompt-version grouping.

Implementation commit: **ac8d805** · Model: **gpt-5.6-luna** · Service tier requested: **Fast** · Report date: **September 16, 2026**

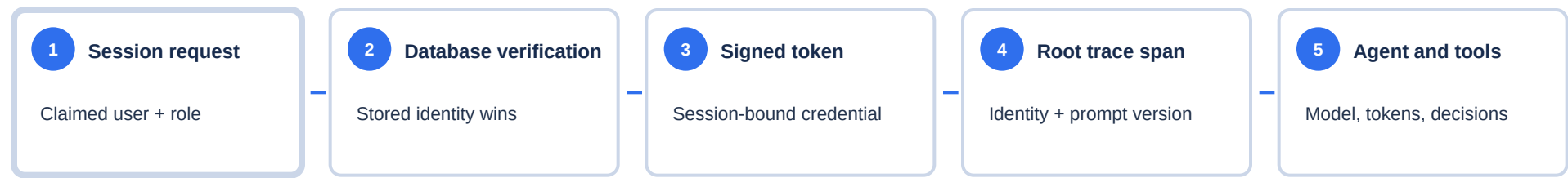
Screenshots in this report were captured from the local self-hosted Langfuse instance and show the actual Homework 2 traces selected during validation.

1. What we implemented

The endpoint establishes identity before model execution, keeps authorization context on the server, and records application attributes on the same OpenTelemetry spans created by the agent instrumentation.



Request flow



Security property: chat text never establishes identity. The server injects the verified AuthContext, and deterministic tools enforce scope before returning data or changing state.

2. Validation and live coverage

Check	Observed result
New offline authentication tests	2 passed
Supplied focused HW2 test	1 passed
Complete offline suite	136 passed; 13 skipped; expected xfail/xpass markers
JSON trace artifact	Exactly 2 objects with all required fields
Prompt restoration	Current hash restored to f4cf9d1978a4

Six requests traced with fresh sessions

Scenario	Role	Main evidence
Shopper order status	shopper / 1	Allowed get_order; accurate delivered status
Out-of-window refund	shopper / 1	Order and policy retrieval chain
Cross-store lookup	merchant / 9002	Denied get_order with recorded reason
Store return policy	merchant / 9002	get_my_store → get_policy; correct 14-day override
Constrained product search	shopper / 1	Filtered and sorted product results
Support order status	support / 9501	Allowed broad support access

All six root spans recorded **cartwheel.session_id**, verified role and user ID, scenario ID, and prompt version. Model spans captured messages and token counts; tool spans captured arguments, structured results, and permission decisions.

3. Authorization held at the tool boundary

Merchant 9002 asked for order 4127 outside store 2. The root trace shows the request, a concise denial, the complete span tree, token and cost totals, and the verified merchant identity with prompt version f4cf9d1978a4.

langfuse

v3.225.8 OSS

Cartwheel Course

Cartwheel Dev

Traces

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Star Langfuse

See the latest releases and help grow the community on GitHub

Langfuse

35k

Settings

Book a call

Support

Student

student@exampl...

Trace

cartwheel.session_message: 8daef78ce9e46d44aae3b7d2f8e08d80

Search

4.68s

\$0.002736

4.68s

\$0.002736

Agent Workflow

4.68s

\$0.002736

cartwheel-support.agent

4.68s

\$0.002736

openai.response

1.82s

802 → 66

(Σ 868)

\$0.001198

get_order

0.00s

openai.response

2.85s

914 → 104

(Σ 1,018)

\$0.001538

Graph

Aggregated

Expanded

Start

cartwheel.session_message

Agent Workflow

cartwheel-support.agent

openai.response (2/2)

get_order

end

cartwheel.session_message :

2026-09-16 16:14:02.441

Latency: 4.68s

Env: default

\$0.002736

1,716 prompt → 170 completion (Σ 1,886)

Preview

Log View

Scores

Formatted

JSON

User

Show me order 4127.

Assistant

I can't show order **4127** because your merchant account doesn't have permission to view it. I can help look up orders belonging to your store instead.

Corrected Output

JSON

Click to add corrected output

Metadata

Path	Value
attributes	5 items
cartwheel.session_id	"25b8a31c-988a-4bde-bf0f-17eef1249dd6"
cartwheel.user_role	"merchant"
cartwheel.user_id	"9002"
cartwheel.prompt_version	"f4cf9d1978a4"
cartwheel.scenario_id	"hw2-live-merchant-denial"
resourceAttributes	5 items
telemetry.sdk.language	"python"
telemetry.sdk.name	"opentelemetry"
telemetry.sdk.version	"1.43.0"
service.instance.id	"07fa03c1-fb25-4d87-8593-0c45a51b72eb"
service.name	"unknown_service"
scope	2 items
name	"cartwheel.server"
attributes	empty object

4. The denial is structured and inspectable

The `get_order` tool received `order_id` 4127 and returned `error=permission_denied` with a precise authorization reason. The trace metadata contains ten attributes, including merchant identity, store ID, and the permission decision.

langfuse

v3.225.8 OSS

Cartwheel Course

Cartwheel Dev

Traces

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Star Langfuse

See the latest releases and help grow the community on GitHub

Langfuse 35k

Settings

Book a call

Support

Student student@exampl...

Trace

cartwheel.session_message: 8daef78ce9e46d44aae3b7d2f8e08d80

get_order :

2026-09-16 16:14:04.261

Latency: 0.00s Env: default

Preview Scores

Formatted JSON

Input

Path	Value
order_id	4127

Output

Path	Value
ok	false
error	"permission_denied"
reason	"role 'merchant' (user 9002) may not view order #4127"

Corrected Output

JSON

Click to add corrected output

Metadata

Path	Value
> attributes	10 items
> resourceAttributes	5 items
> scope	3 items

Graph

Aggregated Expanded

graph TD; start((start)) --> cartwheel_session_message[cartwheel.session_message]; cartwheel_session_message --> agent_workflow[Agent Workflow]; agent_workflow --> cartwheel_support_agent[cartwheel-support.agent]; cartwheel_support_agent --> openai_response_1[openai.response (1/2)]; openai_response_1 --> get_order[get_order]; get_order --> openai_response_2[openai.response (2/2)]; openai_response_2 --> end((end))

5. Authenticated retrieval fixed store ambiguity

The merchant policy trace shows a successful two-tool chain. The final reply names Juniper Home Goods, gives the 14-day delivery-based window, and cites the exact store policy while the root span retains the verified merchant context.

langfuse

v3.225.8 OSS

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Star Langfuse

See the latest releases and help grow the community on GitHub

Langfuse

35k

Settings

Book a call

Support

Student

student@exampl...

Cartwheel Course

Cartwheel Dev

Traces

Trace

cartwheel.session_message: 02ec4fb708d61731f4eee56663395abb

2026-09-16 16:14:07.133

Latency: 7.47s

Env: default

\$0.004269

2,919 prompt → 225 completion (Σ 3,144)

Preview

Log View

Scores

Formatted

JSON

User

What's our store's return window?

Assistant

Juniper Home Goods accepts returns **within 14 days of delivery**. This store-specific window takes precedence over Cartwheel's default 30-day window. (Policy: store-juniper-home-goods-policy)

Corrected Output

JSON

Click to add corrected output

Metadata

Path	Value
attributes	5 items
cartwheel.session_id	"b72b29ae-09eb-4c6b-8c0b-5979051d7b37"
cartwheel.user_role	"merchant"
cartwheel.user_id	"9002"
cartwheel.prompt_version	"f4cf9d1978a4"
cartwheel.scenario_id	"hw2-live-store-policy"
resourceAttributes	5 items
telemetry.sdk.language	"python"
telemetry.sdk.name	"opentelemetry"
telemetry.sdk.version	"1.43.0"
service.instance.id	"07fa03c1-fb25-4d87-8593-0c45a51b72eb"
service.name	"unknown_service"
scope	2 items
name	"cartwheel.server"
attributes	empty object

Graph

Aggregated

Expanded

graph TD; start([start]) --> session_message[cartwheel.session_message]; session_message --> agent_workflow[Agent Workflow]; agent_workflow --> support_agent[cartwheel-support.agent]; support_agent --> openai_response_1[openai.response 1/3]; support_agent --> openai_response_2[openai.response 2/3]; support_agent --> openai_response_3[openai.response 3/3]; openai_response_1 --> get_my_store[get_my_store]; openai_response_2 --> get_policy[get_policy]; openai_response_3 --> end([end]);

6. The exact policy is visible on the tool span

The `get_policy` span shows the exact store policy ID, complete policy body, successful result, `execute_tool` operation, merchant identity, store 2, and `permission_denied=false`.

langfuse v3.225.8 OSS

Cartwheel Course / Cartwheel Dev / Traces

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Star Langfuse

See the latest releases and help grow the community on GitHub

Langfuse 35k

Settings

Book a call

Support

Student student@exampl...

Trace cartwheel.session_message: 02ec4fb708d61731f4eee56663395abb

Search

get_policy :

+ Add to datasets

Annotate

Playground

Add comment

2026-09-16 16:14:11.342

Latency: 0.01s Env: default

Preview

Scores

Formatted JSON

Input

Path Value

policy_id "store-juniper-home-goods-policy"

Output

Path Value

ok true

policy_id "store-juniper-home-goods-policy"

title "Juniper Home Goods store policy"

audience "all"

body "# Juniper Home Goods store policy
Juniper Home Goods accepts returns within 14 days of delivery, instead of the Cartwheel platform default of 30 days. This store window takes precedence for all Juniper Home Goods orders."

Corrected Output

Click to add corrected output

Metadata

Path Value

attributes 9 items

traceloop.span.kind "tool"

gen_ai.tool.name "get_policy"

gen_ai.tool.type "function"

gen_ai.provider.name "openai"

gen_ai.operation.name "execute_tool"

cartwheel.user_role "merchant"

cartwheel.user_id "9002"

cartwheel.store_id "2"

cartwheel.permission_denied false

resourceAttributes 5 items

telemetry.sdk.language "python"

Graph

Aggregated Expanded

__start__

cartwheel.session_message

Agent Workflow

cartwheel-support.agent

openai.response (3/3)

__end__

get_my_store

get_policy

langfuse

v3.225.8 OSS

Cartwheel Course

Cartwheel Dev

Traces

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Star Langfuse

See the latest releases and help grow the community on GitHub

Langfuse 35k

Settings

Book a call

Support

Student student@exampl...

Trace cartwheel.session_message: 3f7b49247ae3390e30eea6fc4bd6c687

Search

Trace

Search

Trace

cartwheel.session_message

4.46s

Σ \$0.002591

cartwheel.session_message

4.46s

Σ \$0.002591

Agent Workflow

4.46s

Σ \$0.002591

cartwheel-support.agent

4.46s

Σ \$0.002591

openai.response

1.61s

738 → 65 (Σ 803)

\$0.001128

get_order

0.00s

openai.response

2.82s

923 → 90 (Σ 1,013)

\$0.001463

2026-09-16 16:18:05.032

Latency: 4.46s

Env: default

\$0.002591

1,661 prompt → 155 completion (Σ 1,816)

Preview

Log View

Scores

User

What's the status of order 4127?

Assistant

Order 4127 from Blue Heron Ceramics was delivered on June 19, 2026. It shipped on June 16, 2026. The order is currently marked eligible for a refund.

Corrected Output

Click to add corrected output

Metadata

Path	Value
attributes	5 items
cartwheel.session_id	"9041a9c9-6e20-436d-8fa7-3838d558248f"
cartwheel.user_role	"shopper"
cartwheel.user_id	"1"
cartwheel.prompt_version	"d108f6949e5e"
cartwheel.scenario_id	"hw2-prompt-before"
resourceAttributes	5 items
telemetry.sdk.language	"python"
telemetry.sdk.name	"opentelemetry"
telemetry.sdk.version	"1.43.0"
service.instance.id	"394027df-0cf7-423d-b5d2-bdc54f9782f3"
service.name	"unknown_service"
scope	2 items
name	"cartwheel.server"
attributes	empty object

Graph

Aggregated

Expanded

start_

cartwheel.session_message

Agent Workflow

cartwheel-support.agent

openai.response (2/2)

get_order

end_

8. Findings and remaining deliverable

Identity is observable

The root span makes the server-verified caller and session inspectable without relying on user-provided chat text.

Authorization is measurable

Allowed and denied tool results share one schema, so permission-denied prevalence can be counted later.

Trace trees explain behavior

The selected traces connect the initial request to model calls, tool arguments, structured results, and final reply.

Prompt experiments are groupable

A template-only hash changed from d108f6949e5e to f4cf9d1978a4 while identity and request conditions remained fixed.

Committed submission files

- observability/instrument.py — authenticated tool-span attributes
- server/app.py — session creation and traced message endpoint
- tests/test_observability.py — offline authentication checks
- hw2-traces.json — two selected completed traces

Remaining manual action

Record one continuous video of no more than five minutes. Run one authentication test, open both selected traces, explain how identity was established, and walk through each tool call and result. Do not mark the video complete until it has been recorded.

Repository state: main is two commits ahead of origin/main. Homework 2 implementation commit: **ac8d805**.